MCP Summary

What MCP Is

Model Context Protocol (MCP) is a system that helps AI models, like chatbots, connect to other tools, apps, and data sources. Normally, AI can only answer questions based on its training data. MCP acts like a bridge to the outside world, letting AI access real-time information or perform tasks.

Think of it like a universal adapter (like USB-C). Instead of creating a new connection for each tool, MCP provides one standard way for AI to connect to many services like databases, APIs, and applications.

MCP has three core components: the **Host** (the AI app the user interacts with, like Claude or ChatGPT), the **Client** (translates requests into a format the server understands), and the **Server** (the external tool or database that returns data or performs actions).

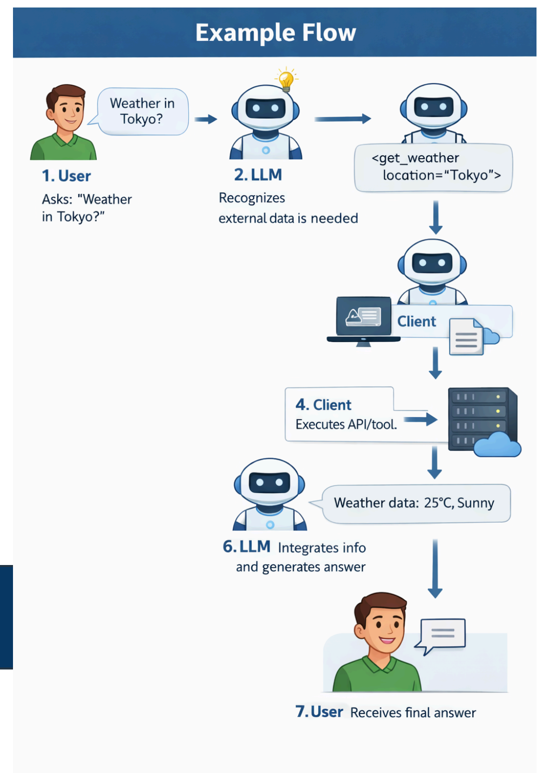

Figure 1: A step-by-step MCP flow where a user asks "Weather in Tokyo?", the LLM recognizes it needs external data, the client calls the weather API, and the result is returned to the user.

How LLMs Use Tools

Large language models (LLMs) can now call tools to fetch data, run calculations, or access APIs. MCP helps standardize this process:

1. **Receive Tool Instructions** – The LLM gets a list of tools with names, descriptions, and input/output formats (e.g., weather API needs a city name).
2. **Recognize Tasks That Require Tools** – AI identifies when it can’t answer with its training data alone (e.g., “Current stock price of Tesla?”).
3. **Generate Tool Call** – The LLM formats a request to the client in MCP’s standard structure.
4. **Client Executes Tool** – The client runs the tool/API and fetches results.
5. **Return and Integrate Results** – The client sends results back, and the LLM integrates them into the final answer.
6. **Optional Fine-Tuning & Safety** – Developers fine-tune tool calls, and humans may approve sensitive requests.